

머신 러닝 - 결정 트리

개와 고양이 분류

In [4]:

```
pip install sklearn
```

Collecting sklearn

Using cached sklearn-0.0.tar.gz (1.1 kB)

Requirement already satisfied: scikit-learn in c:\Users\Wswcom\Anaconda3\lib\site-packages (from sklearn) (0.24.2)

Requirement already satisfied: threadpoolctl>=2.0.0 in c:\Users\Wswcom\Anaconda3\lib\site-packages (from scikit-learn->sklearn) (2.2.0)

Requirement already satisfied: joblib>=0.11 in c:\Users\Wswcom\Anaconda3\lib\site-packages (from scikit-learn->sklearn) (1.1.0)

Requirement already satisfied: numpy>=1.13.3 in c:\Users\Wswcom\Anaconda3\lib\site-packages (from scikit-learn->sklearn) (1.20.3)

Requirement already satisfied: scipy>=0.19.1 in c:\Users\Wswcom\Anaconda3\lib\site-packages (from scikit-learn->sklearn) (1.7.1)

Building wheels for collected packages: sklearn

Building wheel for sklearn (setup.py): started

Building wheel for sklearn (setup.py): finished with status 'done'

Created wheel for sklearn: filename=sklearn-0.0-py2.py3-none-any.whl size=1309 sha256=e9df43ce0e3ff8936557b8ce1ffa1b29cc8da5ff59165278122b563b58b65586

Stored in directory: c:\Users\Wswcom\AppData\Local\Pip\Cache\Wheels\We4W7bW98Wb6466d71b8d738a0c547008b9eb39bf8676d1ff6ca4b22af1c

Successfully built sklearn

Installing collected packages: sklearn

Successfully installed sklearn-0.0

Note: you may need to restart the kernel to use updated packages.

In [1]:

```
from sklearn import tree
```

```
x=[[0,1,0,1,1],[0,1,0,0,0],[1,0,1,0,0]]  
y=[2,2,1]
```

```
c=tree.DecisionTreeClassifier()  
c=c.fit(x,y)
```

In []:

In [6]:

```
c.predict([[1,1,1,0,1]])
```

Out[6]:

```
array([2])
```

In []:

부,모의 키로 자녀의 키가 어떤 그룹에 속할지 예측

- 3: 큰 키, 2: 중간 키, 1: 작은 키

In [1]:

```
from sklearn import tree

parent_height = [ [180,165], [175,160], [180,172], [165,160], [171,152]]
girl_height = [3, 2, 2, 1, 1]

dt = tree.DecisionTreeClassifier()

dt = dt.fit(parent_height, girl_height)
```

In [2]:

```
dt.predict([[175, 153]])
```

Out[2]:

```
array([2])
```

In [3]:

```
dt.predict([[165, 180]])
```

Out[3]:

```
array([1])
```

자동차와 컵 구분하기

In []:

```
from sklearn import tree

x=[[0,0,1,1,1],[1,1,1,1,0],[0,1,1,0,0]]
y=[1,2,2]

c=tree.DecisionTreeClassifier()
c=c.fit(x,y)
```

In []:

```
c.predict([[0,1,1,0,0]])
```

In []:

```
c.predict([[0,1,1,1,0]])
```

초 간단 예제 : 사과와 바나나 분류하기

1. 학습 데이터 생성 과일의 특징을 모양 = 길다/둥글다, 재배기후 = 열대/온대, 색깔=노랑/빨강/연두, 씨 = 없다/있다 특징을 이용해서 사과와 바나나를 구분

In []:

```
# 특징(속성) 데이터
x = [['길다', '열대', '노랑', '없다'], ['둥글다', '온대', '빨강', '있다'],
      ['둥글다', '온대', '노랑', '있다'], ['길다', '열대', '연두', '없다']]
# 정답 데이터
y = ['바나나', '사과', '사과', '바나나']
```

학습 데이터의 수치화

모양 = 길다 : 0 / 둥글다 : 1
재배기후 = 열대 : 0 / 온대 : 1
색깔 = 노랑 : 0 / 빨강 : 1 / 연두 : 2
씨 = 없다 : 0 / 있다 : 1

바나나 : 0 / 사과 : 1

In []:

```
x = [[0, 0, 0, 0], [1, 1, 1, 1], [1, 1, 0, 1], [0, 0, 2, 0] ]  
y = [0, 1, 1, 0]
```

결정트리 모델을 사용해서 머신러닝 학습 모델 생성

In []:

```
from sklearn.tree import DecisionTreeClassifier  
  
#학습 모델 객체 생성  
model = DecisionTreeClassifier()  
  
#학습 데이터를 전달하여 학습 모델 완성  
model.fit(x, y)  
  
#생성된 모델을 사용해서 결과값 예측 : [둥글다, 온대, 연두, (씨가) 있다] 를 전달해서 0  
prd = model.predict([[1, 1, 2, 1]])  
print(prd)
```

건강한 다이어트 예측하기

실전문제등으로 사용하기

In [7]:

```
from sklearn import tree  
x=[[0,0,1,1,1,1],[0,0,1,0,0,1],[1,1,1,1,0,0],[1,1,0,1,0,0],[0,0,0,0,0,1]]  
y=[1,1,0,0,1]  
prd=tree.DecisionTreeClassifier()  
prd=prd.fit(x,y)
```

In [8]:

```
prd.predict([[0,0,1,0,1,0]])
```

Out[8]:

```
array([1])
```

In [14]:

```
prd.predict([[0,1,1,1,1,1]])
```

Out[14]:

```
array([0])
```

In []:

```
%%html
<style type='text/css'>
.CodeMirror{
font-size: 12px;
font-family: D2CODING;
}

div.output_area pre {
font-size: 12px;
font-family: D2CODING;
}
</style>
```

In []:

In []: